

Xingyang (Melody) Bai

Software Engineer | Backend Systems, Accessibility, Data Pipelines, and Databases

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Computer science student building accessible full-stack applications, backend and data systems, and automated testing tools.

◆ EDUCATION

University of Michigan

B.S.E. in Computer Science

Ann Arbor, MI

Expected May 2027

Relevant Coursework: Software Engineering and Automated Testing; Data Structures and Algorithms; Computer Organization

◆ PROJECTS

● Academic Researcher (AP) Radar

July 2026 - Present

Python, Streamlit, PostgreSQL, Web Scraping

- Building Python web-scraping and data pipelines that store structured metrics in PostgreSQL and power real-time Streamlit search and filtering.
- Designing scalable database schemas and update workflows for structured retrieval.

● Accessible Infographic Converter

May 2026 - June 2026

React, Vite, Django, Python

- Built a React/Vite and Django application with secure file uploads that converts inaccessible flowcharts and timelines into screen-reader-compatible PDFs.

● Automated Testing and Defect Detection Suite

March 2026 - April 2026

Python, C++, AWS

- Built Python/C++ automated-testing, mutation-testing, and debugging tools on AWS to isolate defects and evaluate software reliability.

● Social Media Database System

March 2026

Java, JDBC, SQL

- Designed a Java/JDBC relational database with DDL constraints and optimized SQL queries for reliable large-scale ingestion and retrieval.

● Relational Database Engine

June 2025

C++

- Built an in-memory C++ database engine with custom SQL parsing, hash/BST indexing, edge-case testing, and memory-leak validation.

◆ EXPERIENCE

● Dr. Maldonado's Research Group - Web Developer

September 2024 - May 2025

Ann Arbor, MI

- Automated the migration of 1,000+ pages with Python and redesigned an electrochemistry website in HTML5, CSS, and JavaScript to improve accessibility, navigation, and consistency.

● Shapiro Undergraduate Library - Accessibility Ambassador

January 2025 - Present

Ann Arbor, MI

- Created 10+ tutorials for Adobe Acrobat, NVDA, Read&Write, and ZoomText, informed by student and staff interviews identifying accessibility barriers.

◆ TECHNICAL SKILLS

Programming Languages: Python, C++, C, Java, SQL, JavaScript, HTML5/CSS, LaTeX

Frameworks, Databases, and Cloud: Django, React, Vite, Streamlit, Pandas, PostgreSQL, JDBC, AWS

Developer Tools and Concepts: Git, Linux/Unix, backend development, relational database design, data pipelines, web scraping, automated testing, mutation testing, debugging, database indexing